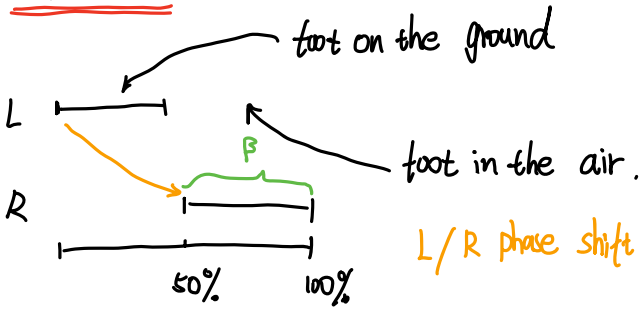
口试课程，难度中上，教很多仿真的 动力学 + 控制 理论，
很有意思。

1、Gaits



L/R phase shift

Duty factor.

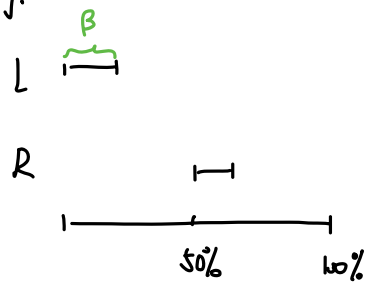
β: % each foot on the ground.

symmetrical Gaits 可以由 β 完全定义。

non-symmetrical Gaits 需要更多来定义

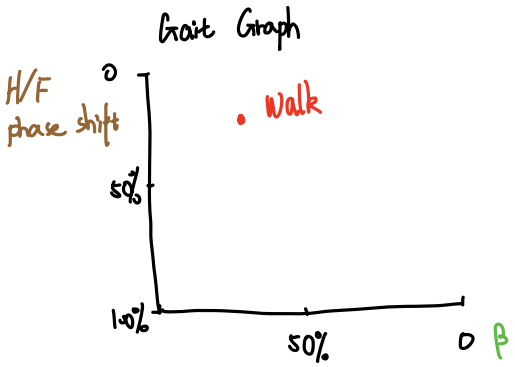
symmetrical：
左右脚对称。

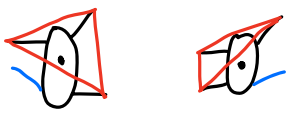
e.g.



4条腿走路：

- H/F phase shift.



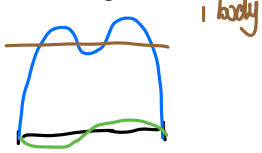


前脚迈出之后，后脚迈到侧更稳

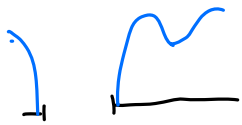
后脚迈出之后，前脚迈同侧更稳。

GRF (Ground Reaction Forces).

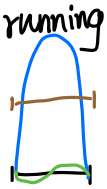
Walking



around 1 body weight



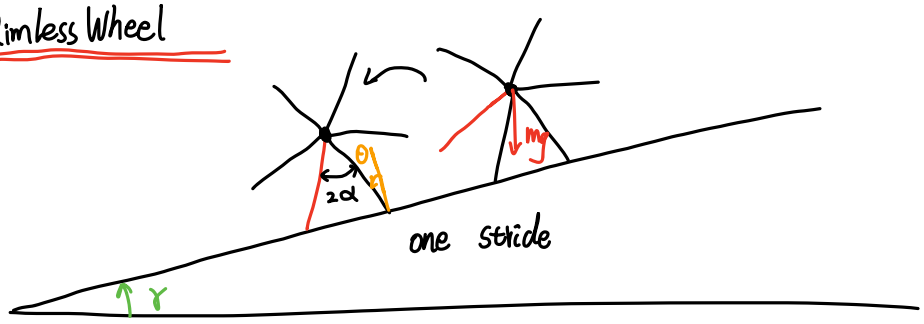
running



around 2 body weight.

....

2. Rimless Wheel



① equition of motion in one stride

$f(x)$

② event function $e(x)$ for the stopping time of ODE.



③ reaction function / collision function $g(x)$

$\Rightarrow$ compute θ_{i+1} by θ_i

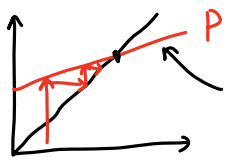
↳ initial condition for ODE to start again.

$\Rightarrow \theta_{i+1} = P(\theta_i)$ Poincaré map. discret time dynamics.

↳ $\begin{bmatrix} \theta_i \\ \dot{\theta}_i \end{bmatrix}$

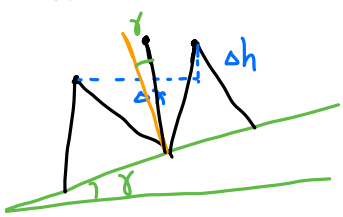
Autonomous System.

↳ fix θ_i to draw a cobweb.



$|slop| < 1$ → stable fixed point.

Energy Economy.



$COT = \dfrac{E}{\Delta x\, mg}$ $\quad E = \Delta h\, mg \Rightarrow COT = \dfrac{\Delta h}{\Delta x}$.

Cost of transport.

(the energy that needed, to do a unit transport)

Human : COT around 1.

3. Scalling & Normalization.

use relevant values to define units.

[l] leg length

[m] body mass. $\Longrightarrow$

[g] gravity

Force $[mg]$

Time $\left[\sqrt{\frac{l}{g}}\right]$ time to fall half a leg length.

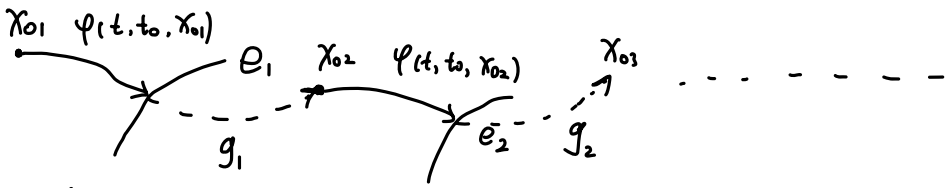
Frequency $\left[\sqrt{\frac{g}{l}}\right]$

Speed $[\sqrt{gl}]$

4. Modelling Hybrid Dynamic Systems

Structure.

$\dot{x} = f(x) \longrightarrow x(t) = \varphi(t, t_0, x_0)$



↳ EOM

$M\ddot{\vec{q}} = f_{cg}(\vec{q}, \dot{\vec{q}}) + \tau$ $\xrightarrow[L = T - V]{\text{Lagrange equations}}$ $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \tau$ → f_{np}.

contact force λ $\quad J^T\vec{\lambda} \longrightarrow \vec{V} = J\dot{\vec{q}}$ (DMS)

功率 : Power

$P_c = \dot{\vec{c}}^T\vec{\lambda} = \dot{\vec{q}}^T J^T\vec{\lambda} = \dot{\vec{q}}^T\vec{\tau}_\lambda = P_q$

$J^T\vec{\lambda} = \vec{\tau}_\lambda$

$M\ddot{\vec{q}} = f_{cg}(\vec{q}, \dot{\vec{q}}) + \tau + \tau_\lambda$ → put the contact constraints inside.

constraints : $\vec{C}(\vec{q}) = 0$.

$\dot{\vec{C}}(\vec{q}, \dot{\vec{q}}) = \frac{\partial \vec{C}}{\partial \vec{q}}\dot{\vec{q}} = J(\vec{q})\dot{\vec{q}} = 0$

$\ddot{\vec{C}} = J\ddot{\vec{q}} + \dot{J}\dot{\vec{q}} = 0$

collision : constraints changed.

in DCLL $\dot{\vec{C}} = 0 = J\dot{\vec{q}}$

Problem with Event-Base :

① one specific event \ event with order.

② Zeno Behavior : never stop . need ∞ stride to stop.

③ Sliding and Friction

5. Limit Cycles

$\vec{x}^{k+1} = P(\vec{x}^k)$ period time T.

fixed point $P(\vec{x}^*) = x^*$

not all x need to be periodic.  $\xrightarrow{x}$ not periodic.

① solve the fixed point with like (Newton - Raphson)

② Stability.

$J = \left.\frac{\partial P}{\partial x}\right|_{x^*}$, eigenvalue $|\lambda| < 1$.

$\lambda = 0$ disturbance along the flow.

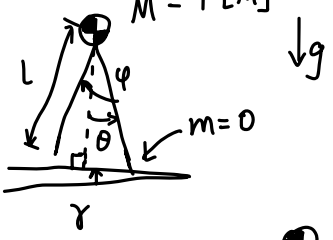
$\lambda = 0$. collision.

Compute J.

① △ differential.

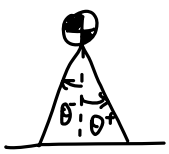
② with Φ.

6. The Simplest Walking Model



$m = \beta[M] \longrightarrow \beta \to 0$.

$\Rightarrow f(x)$



$\varphi - 2\theta = 0$
$\theta < 0$
$\Big\} e(x)$.

$x^- \xrightarrow{g(x)} x^+$

7. SLIP - Model



$\vec{q} = \begin{pmatrix} x \\ y \end{pmatrix}$, phase $\{$ flight^1 stance flight$^2\}$.

x_{cp}. contact point.

$r = \begin{pmatrix} x - x_{cp} \\ y - 0 \end{pmatrix}$

$l = \| \vec{r} \|$

$F_{spring} = k(l_0 - l)$

$\Longrightarrow$ $f(x)$ Flight

$$\ddot{\vec{q}} = \begin{pmatrix} \ddot{x} \\ \ddot{y} \end{pmatrix} = \begin{pmatrix} 0 \\ -g \end{pmatrix}$$

$f(x)$ stance

$$\ddot{\vec{q}} = \begin{pmatrix} 0 \\ -g \end{pmatrix} + \frac{1}{m_0} F_{spring} \frac{\vec{r}}{l}$$

Jump - Set. $e(x)$.

$e_1^2 = -(y - l_0 \cos\alpha)$ touch down

$e_2^3 = (l - l_0)$ lift off

$e_3^1 = -\dot{y}$ apex change.

Jump - Map $g(x)$

touch down.

$\vec{q}^+ = \vec{q}^-$

$\dot{\vec{q}}^+ = \dot{\vec{q}}^-$

$x_{cp}^+ = x^- + l_0 \sin\alpha$

lift off

$\vec{q}^+ = \vec{q}^-$

$\dot{\vec{q}}^+ = \dot{\vec{q}}^-$

$x_{cp}^+ = x_{cp}^-$

apex change

$\vec{q}^+ = \vec{q}^-$

$\dot{\vec{q}}^+ = \dot{\vec{q}}^-$

$x_{cp}^+ = x_{cp}^-$

$\Longrightarrow P(\underline{x})$ $\underline{x} = \begin{bmatrix} y \\ \dot{x} \\ \dot{y} \end{bmatrix}$ x is not periodic.

EV: $\lambda_1 = 1$ $\lambda_2 = 0.72$ $\lambda_3 = 0$.

$\lambda_3 = 0$: disturbances along the flow

$\lambda_2 = 0.72$: orthogonal to the flow

$\lambda_1 = 1$: energetically conservative.

points to a different energy level: Bifurcation

Continuation of Gaits.

8. Continuation of Gaits

$\vec{r}(\vec{x}, \gamma) = \vec{0}$. → find solution family.

start with one known solution:

$\vec{x}_k, \gamma_k$ with $\vec{r}(\vec{x}_k, \gamma_k) = 0$.

we have $d\gamma_k$ $\quad d\gamma = \frac{\partial r}{\partial \vec{x}} d\vec{x} + \frac{\partial r}{\partial \gamma} d\gamma \overset{!}{=} 0$.

$$d\vec{x} = -\left(\frac{\partial r}{\partial \vec{x}}\right)^{-1} \frac{\partial r}{\partial \gamma} d\gamma$$

fails at turning point and bifurcation

$\vec{x}_p = \vec{x}_k + d\vec{x}$

$\gamma_p = \gamma_k + d\gamma$ $\quad$ if $r(\vec{x}_p, \gamma_p) = 0$ $\quad \vec{x}_{k+1} = \vec{x}_p$

为了越过 turning point.

treat all variables equally $\quad r(\vec{x}, \gamma) = r(\vec{u}) \quad \vec{u} = \begin{pmatrix} \vec{x} \\ \gamma \end{pmatrix}$

find tangent vector $\vec{t}$

$$\left(\frac{\partial r}{\partial \vec{u}}\right)\vec{t} = R\vec{t} = \vec{0}$$

$\|\vec{t}\| = 1$, $\quad \det\left(\begin{bmatrix} R \\ \vec{t}^T \end{bmatrix}\right) > 0$.

$r_{i+1} - r_i = R\Delta u + O^2(u)$

$r_{i+1} \overset{!}{=} 0 \Rightarrow \Delta u = -\text{pinv}(R)\, r_i$

→ with Newton's Method

$u_p = u_k + \vec{t} \cdot h \cdot d$ → Correction

detect bif.: $\vec{t}_k \vec{t}_{k+1} < 0$. (flip) / $\det\left(\begin{bmatrix} R \\ \vec{t}^T \end{bmatrix}\right) = 0$

$\bar{E}$: energy level.

$\Rightarrow$ SLIP-Model

Modify Dynamics

$$r(\vec{x}, s, \bar{E}) = \begin{pmatrix} P(\vec{x}, s) - \vec{x} \\ E(x) - \bar{E} \end{pmatrix}$$

$$\dot{\vec{x}} = \underbrace{f(\vec{x}) + s\nabla E(\vec{x})}_{P(x, s)}$$

9. Time Stepping Methods

Take the "integrated EOMs"

$$\int_{t_A}^{t_E} M\ddot{\vec{q}} - \vec{f}_{cg} - J^T\vec{\lambda} - \vec{\tau}\ dt = 0$$

we can approximate M, f_{cg}, J & $\vec{\tau}$ to be constant during the step.

the only thing we don't know is $\boxed{\int_{t_A}^{t_E} \vec{\lambda}\ dt = \vec{\Lambda}}$

$$M(\dot{\vec{q}}_E - \dot{\vec{q}}_A) - (f_{cg} + \tau)\int_{t_A}^{t_E} 1\ dt - J^T\int_{t_A}^{t_E}\vec{\lambda}\ dt = 0$$

$$\dot{\vec{q}}_E = \cdots\cdots$$

$$\vec{q}_E = \vec{q}_A + \frac{1}{2}(\dot{\vec{q}}_A\Delta t + \dot{\vec{q}}_E\Delta t)$$

How to know $\vec{\Lambda}$ $\Longrightarrow$ Modelling Friction

consider different case to solve → only the relationship between Λ_N and Λ_T.

$$\vec{\Lambda} = \begin{bmatrix}\Lambda_N\\ \Lambda_T\end{bmatrix} \qquad \Lambda_T \in [-\mu\Lambda_N,\ \mu\Lambda_N]$$

↳ We split $J^T\vec{\Lambda} = J_N^T\vec{\Lambda}_N + J_T^T\vec{\Lambda}_T$

take the initial guess $\vec{\Lambda}_N^k$, $\vec{\Lambda}_T^k$ ⟶ We can get $\dot{\vec{q}}_E^{\,k}$

then use the Proximal Point Function.

$$\vec{\Lambda}_N^{k+1} = \text{prox}_{R_0^+}(\vec{\Lambda}_N^k - rJ_N\dot{\vec{q}}_E^{\,k})$$

$$\vec{\Lambda}_T^{k+1} = \text{prox}_{[-\mu\Lambda_N,\ \mu\Lambda_N]}(\vec{\Lambda}_T^k - rJ_T\dot{\vec{q}}_E^{\,k})$$

until they converge ⟶ $\dot{\vec{q}}_E$

because $\dot{C}_E = J\dot{\vec{q}}_E$

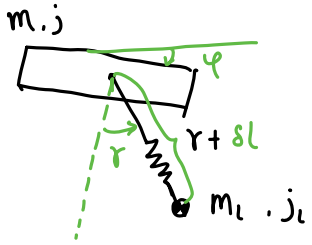
接触速度.

$\dot{C}_E < 0 \Longrightarrow$ 力大 Λ

$\dot{C}_E > 0 \Longrightarrow$ 减小 $\Lambda \longrightarrow 0$ 置0.

$\dot{C}_E = 0 \Longrightarrow$ 保持.

10. The Raibert Hopper



What can we control ?

① Hopping Height (δl , k , F_T , E_k)

② Main Body Pitch (φ)

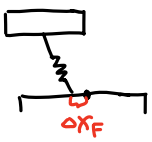
③ Forward Speed & Position (Neutral Point , Δx_F , $\dot{x}$)
(Leg Angle γ)

Flight : Leg Angle γ → Forward Position

need a γ_d . this can derive from x_F (Neutral Point)

first control $\Delta x_F = K(\dot{x}_d - \dot{x})$

then $x_F = l_0 \sin(\gamma + \varphi)$ $\Rightarrow$ γ_d



$x_F = \frac{1}{2}\dot{x} T_s + \Delta x_F$

Landing : Doing Nothing

Compression :

Main Body control . φ . $\varphi_d = 0$

* . $T_\gamma = -T_\varphi$.

Thrust :

Main Body control . φ . $\varphi_d = 0$

* . $T_\gamma = -T_\varphi$.

Hopping Height . δl .

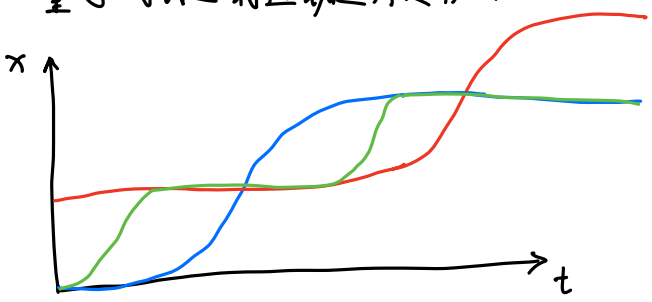
$\delta l = \delta l$ + some positive value .

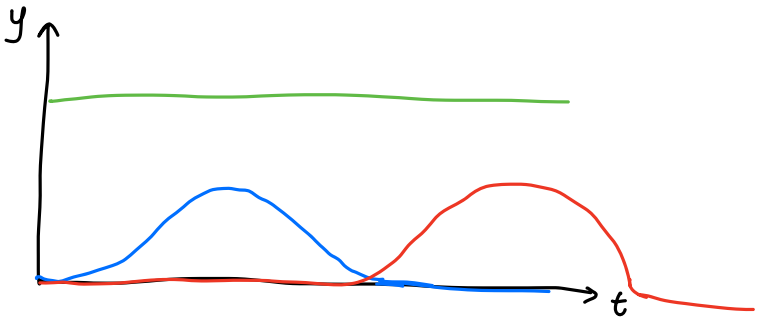
Unloading : Doing Nothing

11. Statically Stable Kinematic Control

Foot Motion Relative to COG.

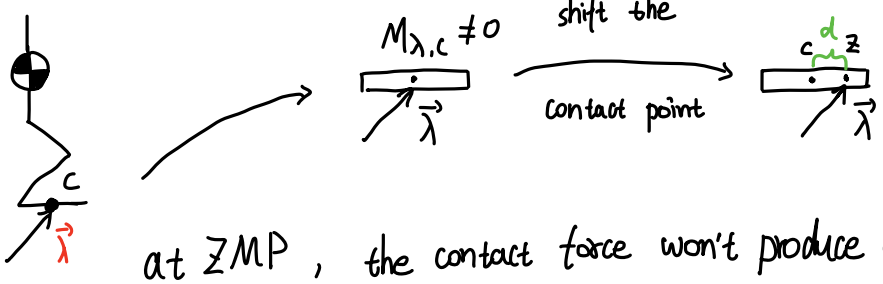
重心与脚的运动是有关联的.





Control: assume COG at hip.
inverse kinematics
PID control on joint level.

12. ZMP - Control



$M_{\lambda,c} = M_{\lambda,z} + d\lambda_y$, $M_{\lambda,z} \overset{!}{=} 0$

at ZMP, the contact force won't produce any Moment that makes the Robot falling down.

① Plan the trajectory of ZMP and Let the foot be there.

② Compute the main body posintion from ZMP with (Linear Inversed Pendulum)

$$z_x = x - \frac{h}{g}\ddot{x} \qquad z_y = y - \frac{h}{g}\ddot{y}$$

③ then compute joint angle_des with inverse dynamics.

④ PID control for joint level.

How to plan trajectory of ZMP?

Optimal Control with input $\dddot{x}$

$z_{ref} = x_{COG}$.

(We already have Dynamics)

$$\dot{\hat{x}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\hat{x} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\dddot{x}$$

$$z_x = (1, 0, -\frac{h}{g})\hat{x}$$

discrete: $A = \begin{bmatrix} 1 & T & \frac{1}{2}T^2 \\ 0 & 1 & T \\ 0 & 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} \frac{1}{6}T^3 \\ \frac{1}{2}T^2 \\ T \end{bmatrix}$ $C = \begin{bmatrix} 1 & 0 & -\frac{h}{g} \end{bmatrix}$

$$\min \sum_{i=k}^{k+N-1} \frac{1}{2}Q(\hat{z}_{i+1} - \hat{z}_{i+1}^{ref})^2 + \frac{1}{2}R\dddot{x}_i^2$$

s.t. $\hat{x}_{k+1} = A\hat{x}_k + B\dddot{x}_k$

$\hat{z}_k = C\hat{x}_k$

$\hat{x}_k = \begin{bmatrix} x_k \\ \dot{x}_k \\ \ddot{x}_k \end{bmatrix}$

13. Virtual Model control (VMC)

Hybrid dynamics.

$$M\ddot{\vec{q}} = f_{cg} + \underbrace{J^T\vec{\lambda} + \vec{\tau}}$$

$$\vec{\tau}_{des} = (f_x \; f_y \; f_\theta \; 0 \; 0 \; \cdots)^T$$

split $J^T = \begin{bmatrix} J_P^T \\ J_A^T \end{bmatrix}$

$$\Longrightarrow \vec{\tau}_{des} = \begin{bmatrix} f_x \\ f_y \\ f_\theta \\ 0 \\ \vdots \end{bmatrix} = \begin{bmatrix} J_P^T \\ J_A^T \end{bmatrix} \vec{\lambda}_{des} + \vec{\tau} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ \tau_1 \\ \vdots \\ \tau_n \end{bmatrix} \rightarrow \text{joint level.}$$

We can control: $f_x = P_x(x_{des} - x) + D_x(\dot{x}_{des} - \dot{x})$

$f_y = P_y - \cdots + D_y - - -$

$f_\theta = P_\theta \cdots + D_\theta - - -$

from $\begin{bmatrix} f_x \\ f_y \\ f_\theta \end{bmatrix} = J_P^T \vec{\lambda}_{des} \longrightarrow \vec{\lambda}_{des}$

then $\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = J_A^T \vec{\lambda}_{des} + \begin{bmatrix} \tau_1 \\ \vdots \\ \tau_n \end{bmatrix} \longrightarrow \vec{\tau}$

from the help of $\vec{\lambda}$ and $\vec{\tau}$, we can do Main Body control.

double stance $\vec{\lambda}_{des} \in \mathbb{R}^4$, We need $\text{pinv}(J_P^T)$, no unique solution.

single stance $\vec{\lambda}_{des} \in \mathbb{R}^2$, no solution for $\begin{bmatrix} f_x \\ f_y \\ f_\theta \end{bmatrix}$.

but we can only take two of them.

e.g. $\begin{bmatrix} f_y \\ f_\theta \end{bmatrix}$, and $(\tilde{J}_P^T)^{-1}$.

ISSUES:
- Dynamics are only amended, not controlled.
- No formal proof of stability
- Different single \ double stance.

14. HZD - Control

Define virtual constraint : $\vec{y} = h(\vec{q}) = 0 \Rightarrow \dot{\vec{y}} = \frac{\partial h}{\partial q}\dot{\vec{q}} = J\dot{\vec{q}} = 0$

$\ddot{\vec{y}} = J\ddot{\vec{q}} + \dot{J}\dot{\vec{q}} = 0$

put this in Dynamics $M\ddot{\vec{q}} = \vec{f}_{cg} + S\vec{\tau}$

then we will have $\vec{\tau}$ = ------ → 使 $\ddot{\vec{y}} = 0$ 的 τ.

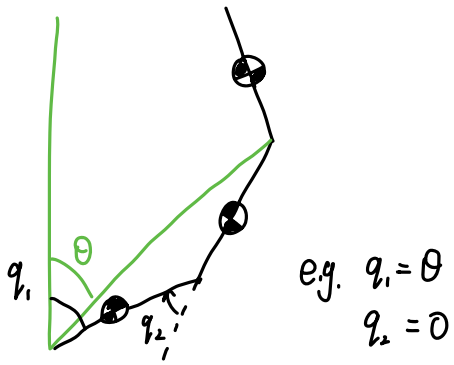
To design the constraints :

- Define a θ (strictly grows monotonic)
  (can not directly control)

we have $\theta = C^T\vec{q}$

- Define the desired joint position $\vec{q}_a = S^T\vec{q}$

$\vec{q}_a \overset{!}{=} h_d(\theta)$



e.g. $q_1 = \theta$
$q_2 = 0$

$\Rightarrow$ constraints : $\vec{y} = h(\vec{q}) = \vec{q}_a - h_d(\theta)$.

desired joint position : Main Body upright.
Stance knee straight
mirrors legs
- - - -

$\min \int \tau^T\tau \, dt$
. - - - - .

collision ?

support change ?

it works, but we need to think about it.

if we take the initial condition fulfill $y = 0$, it works. or

take $\ddot{\vec{y}} = -P\vec{y} - D\dot{\vec{y}}$, to get $\vec{\tau}$

Last thing : Stabilizing the HZD.

① 能量会损耗. $\Rightarrow \dot{\vec{q}}^+ \neq \dot{\vec{q}}^-$
(by collision)



给 Main Body
一个小obtuse角.

② 脚会滑地.

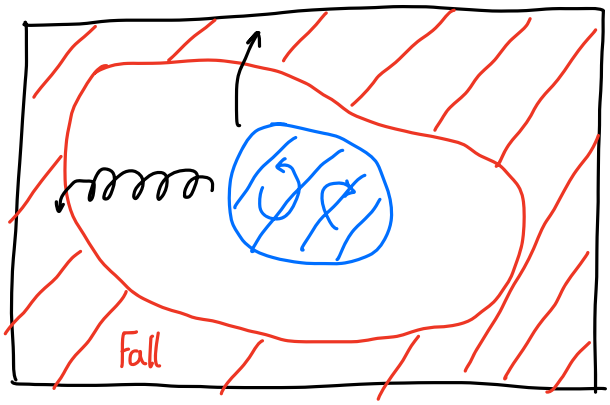


Swing leg touch the ground.

- Extend stance leg
- give a β_{offset}, don't let the leg always straight.

15. Capture Points

- **How to Quantify:**



- Limit Cycle Stability

  $\max.(\text{eig}\|J\|) < 1$, with $J = \frac{\partial P}{\partial \vec{x}}$

  local measure, motion must be periodic

- Static Stability

  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \vec{x}$ (COG) within support polygon.

- ZMP Stability

  $\vec{z}$ within support polygon.

- **Viability Kernel:**

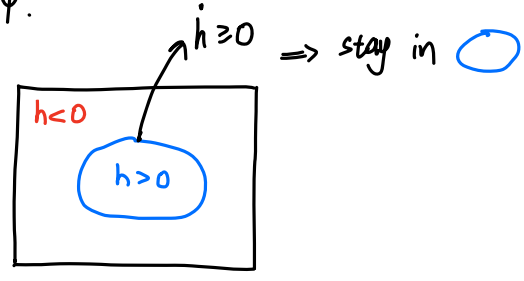
  Controlled Dynamics $\dot{\vec{x}} = f(\vec{x}, \vec{u})$ with controller $\vec{u}$

- Control-Lyapunov functions $V(\vec{x}) \geq 0$

  for $\forall \vec{x}$, there $\exists \vec{u}$, such that $\dot{V}(x,u) \leq 0$.

  $V_\alpha = \{\vec{x} \mid V(\vec{x}) < \alpha\}$

  $\Rightarrow$ find V & α, $\longrightarrow$ max. V_α and $V_\alpha \cap F = \phi$.



- Control-Barrier functions $h(\vec{x})$

  given a function $h(\vec{x})$, we define

  $C = \{\vec{x} \mid h(\vec{x}) \geq 0\}$ C is invariant if $\forall \vec{x} \in \partial C$,

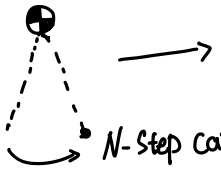
  $\partial C = \{\vec{x} \mid h(\vec{x}) = 0\}$ $\exists \vec{u}$, such that $\dot{h}(\vec{x}, \vec{u}) \geq 0$

- Capturability

  N-step capturable.

  a state $\vec{x}_0$ is N-step capturable if there exist a future input $\vec{u}$,

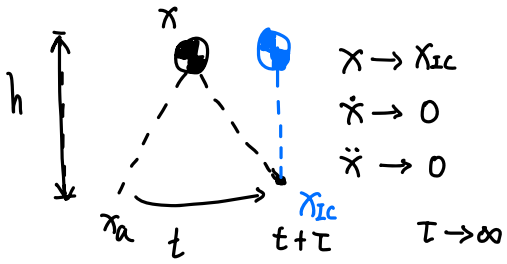
  that prevent falling without taking more than N steps.



→ it becomes N-1 step capturable.

what if step to 1 step capture point ? ⟹ it will stop.

instantaneous capture point. x_{IC}



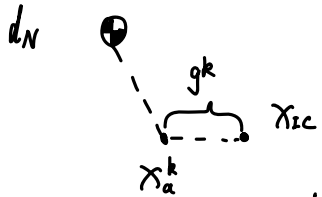
$$x_a = x - \frac{h}{g}\ddot{x} \longrightarrow \frac{h}{g}\ddot{x}(t+\tau) = x(t+\tau) - x_{IC}.$$

⟹ Dynamics of x_{IC}

$$\frac{1}{2}\dot{x}^2 - \frac{1}{2}(x - x_{IC})^2 = 0 \quad (\tau \to \infty)$$

$$\Rightarrow \begin{cases} x_{IC} = x + \dot{x} \\ \dot{x}_{IC} = \dot{x} + \ddot{x} \end{cases} \longrightarrow \boxed{\begin{aligned} x_{IC}(t) &= (x_{IC}(0) - x_a)e^t + x_a \\ \dot{x}_{IC} &= x_{IC} - x_a \end{aligned}}$$

- N-step Capture Region.



$g^k \le d_N \Rightarrow$ N step capturable.

$d_0 = 0$. (stop at 0 step Capture point)

$x_a^{k+1} - x_a^k \le l_{max}$ (max. step length)

$x_{IC}(0) - x_a^k \le d_N$.

$x_{IC}(t_s) - x_a^{k+1} \le d_{N-1}$

We know the dynamics of x_{IC}. $\Rightarrow x_{IC}(t_s) = f(x_{IC}(0), x_a^k)$

$\Rightarrow d_N$ is a function of d_{N-1} and l_{max}

$$d_N = (d_{N-1} + l_{max})e^{-t_s}, \quad d_0 = 0$$

$$\to \infty \quad d_\infty = l_{max}\frac{e^{-t_s}}{1 - e^{-t_s}}$$

16. Optimization

We have a lot of controllers? Which one is the best?

We don't choose them manually, we design a cost function and take controlled dynamics as constraints, give controller a bound.

$$\Rightarrow \min_{T,\ \vec{x}(\cdot),\ \vec{u}(\cdot)} \int_0^T C_L(\vec{x}(t), \vec{u}(t))\,dt + C_M(\vec{x}(T), u(T), T)$$

periodic: $\vec{x}^*(T) - \vec{x}^*(0) = 0$

forward speed: $\vec{x}^*(T) - \vec{x}^*(0) = \vec{v}_{des}\,T$.

(Hybrid) Dynamics: $\dot{\vec{x}}_i(t) = f_i(\vec{x}, \vec{u})$

$$x_{i+1} = g_i^{i+1}(\vec{x}(t_i))$$

$$e_i^{i+1}(\vec{x}(t_i)) = 0 \quad , \quad \dot{e}_i^{i+1}(x(t_i)) > 0$$

Bound: $u_{min} \le u \le u_{max}$

$$x_{min} \le x \le x_{max}.$$

$$t_{min} \le T \le t_{max}.$$

contact force? other constraints?

How to solve? How to deal with the constraints?

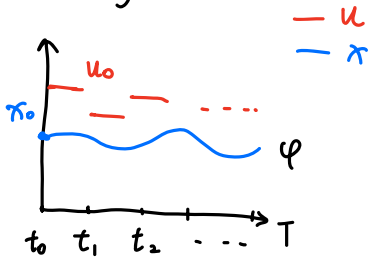
a) Single shooting

devide T to N part.

initial guess $u_0 = c$ in $t_0 \sim t_1$

$x(t_0) = x_0$

$\Rightarrow$ traj. φ

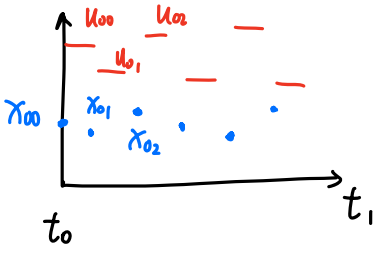


b) multiple shooting.

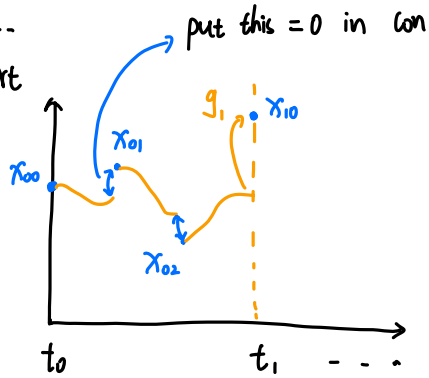
intial guess $u_{00}, u_{01}, \cdots$

$x_{00}, x_{01}, \cdots$

guess the traj. of first part



put this = 0 in constraints.



c) Direct Collocation.

numerical time stepping.

- Forward Euler $x_{i+1} = x_i + h\dot{x}_i$
- Backward Euler $x_{i+1} = x_i + h\dot{x}_{i+1}$
- Central Difference $x_{i+1} = 2x_i - x_{i-1} + h^2\ddot{x}_i$
- Newmark-β - - - - - -
- Velocity-Verlet. - - - - - -

need very short $h = t_{i+1} - t_i$.

17. Energetics of Locomotion

The cost of Transport

$$COT = \frac{E}{\Delta x\, mg}$$ How to define E?

Measures of Energy

- $E = E_{fuel}$. (gas, battery, foodstuff).

  Gravity Energy.

  $= mg\Delta h \Longrightarrow COT = \frac{\Delta h}{\Delta x}$.

- $E = W_{mech}$ (mechanical work)

  $= \int_0^T \vec{\tau}\dot{\vec{q}}\, dt = W^+_{mech} + W^-_{mech}$

- $E = W_{loss}$

  $= \frac{1}{k}\int_0^T \vec{\tau}^T\tau\, dt$

- $E = W_{el} = W_{loss} + W_{mech}$.

take $E = \frac{W^+}{\eta^+} + \frac{W^-}{\eta^-}$

$W^+ > 0 \quad W^- < 0$

perfect work, perfect recovery.

$\Longrightarrow \eta^+ = 100\%, \ \eta^- = 100\%$.

Human: $\eta^+ = 25\%, \ \eta^- = -120\%$.

What produce W?

Environment? $= 0$

Conversion loss and Friction ✓

Breaking ✓ (W^-).

↳ take COT as a cost function, to build a Optimal Control problem.

$$\min_{q, \dot{q}, u, T} COT$$

s.t. hybrid dynamic

bounds.

$\Longrightarrow$ Optimal Motions.